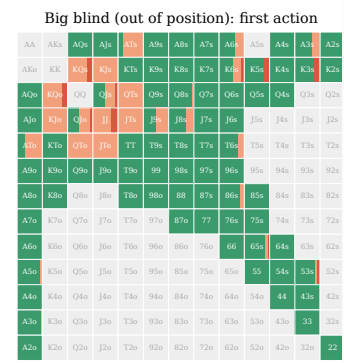
Solving Poker

The numerical analysis behind poker solvers
— and a second-order finish for them

Numerical Analysis Research Club

POKER Ten years of superhuman poker

- **2015** Cepheus: heads-up *limit* hold'em "essentially weakly solved" with CFR$^+$
  Bowling, Burch, Johanson, Tammelin, *Science*
- **2017** DeepStack beats 33 professionals at heads-up *no-limit* (44,852 hands, ≈490 mbb/hand)
  Moravčík et al., *Science*
- **2017** Libratus beats four top pros over 120,000 hands (14.7 big blinds / 100 hands)
  Brown & Sandholm, *Science* 2018
- **2019** Pluribus: six-player no-limit
  Brown & Sandholm, *Science*
- **Today**: every serious player studies with solvers.



A solver's answer (we computed): how the big blind should play every hand on one river.

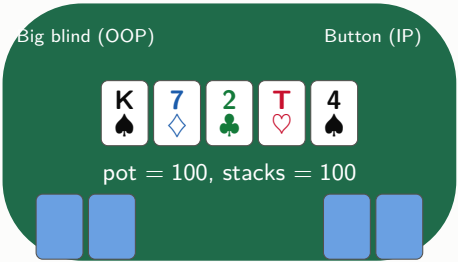
The plan: poker $\leftrightarrow$ math



- The equilibrium of a poker game is the solution of a (huge, structured) bilinear saddle-point problem.
- Solvers use a *first-order*, matrix-free iteration: cheap steps, slow tail.
- Our contribution: treat the end of the solve as a *nonsmooth equation* and finish with a proximal semismooth Newton method, warm-started by the solver.

POKER Texas hold'em in one slide

- Two players, each dealt two private cards.
- Five public *board* cards revealed in rounds: flop (3), turn (1), river (1).
- A betting round before and after each reveal: *check, bet, call, raise, fold*.
- *No-limit*: any bet size up to your stack.
- Best five-card hand from seven cards wins at showdown — unless everyone else folded.



Our running example: the river, last betting round.

The big blind acts first ("out of position").

Two-player zero-sum games and Nash equilibrium

Strategies $x \in X$, $y \in Y$ (compact convex), payoff to player 1: bilinear $u(x, y) = x^\top A y$.

Minimax theorem (von Neumann 1928)

$$\max_{x \in X} \min_{y \in Y} x^\top A y \;=\; \min_{y \in Y} \max_{x \in X} x^\top A y \;=:\; v^* \quad \text{(the value of the game)}.$$

A pair (x^*, y^*) attaining both sides is a **Nash equilibrium**.

The error metric: exploitability

$$\mathrm{expl}(x, y) \;=\; \tfrac{1}{2}\Big(\max_{x'} x'^\top A y \;-\; \min_{y'} x^\top A y' \Big) \;\geq 0, \qquad \mathrm{expl}(x, y) = 0 \iff \text{equilibrium}.$$

- Playing x^* guarantees $\geq v^*$ against *anyone*: unexploitable, "game-theory optimal".
- expl is a duality gap: the residual every solver reports. Units: chips per hand.

POKER Why equilibria mix: bluffing and pot odds

Toy river: pot 1, bettor has either the nuts or nothing; bets s; caller holds a bluff-catcher.

- Caller indifferent $\Rightarrow$ bettor's bluffs are a fraction

$$\beta = \frac{s}{1+2s} \quad \text{of the betting range.}$$

- Bettor's bluffs indifferent $\Rightarrow$ caller defends with frequency

$$\text{MDF} = \frac{1}{1+s} \quad (\text{“minimum defence frequency”}).$$

- Pure strategies are exploitable; equilibrium *randomises* exactly at indifference.

bet s	bluffs β	MDF
1/3	20%	75%
3/4	30%	57%
1	33%	50%
2	40%	33%

Indifference conditions = complementary slackness.

(We will see the solver reproduce the MDF column.)

MATH Kuhn poker: the smallest interesting example

- Deck  ; each player antes 1, gets one card.
- One bet of size 1; actions check/bet, then fold/call.
- Equilibria form a *one-parameter family* (Kuhn 1950): first player
  - bets J with prob. $\alpha \in [0, \frac{1}{3}]$ (a bluff),
  - bets K with prob. 3α (slow-plays the rest),
  - value $v^* = -\frac{1}{18}$.

We computed (Kuhn)

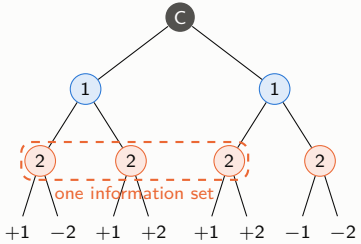
prox-Newton output: $\alpha = 0.198$, K bets $0.593 = 3\alpha$
value $-0.0555\ldots = -1/18$
exploitability 1.4×10^{-16}

The equilibrium set is a **polytope**, not a point: any method must cope with *non-isolated* solutions.

Extensive-form games

- Game tree: chance nodes (deals), decision nodes, terminal payoffs.
- **Information set** I: histories a player cannot distinguish (same public actions, opponent's cards unknown).
- Behavioural strategy: a distribution $\sigma(I) \in \Delta(A(I))$ at every information set.
- Perfect recall: you never forget what you knew or did.

game	information sets
Kuhn	12
Leduc hold'em	288
HU limit hold'em	3.2×10^{14}
HU no-limit hold'em	$\sim 10^{161}$



Player 2 sees the action, not the deal.

MATH The sequence form: equilibrium is a linear program

Sequences of player 1: paths of her own actions; *realisation plan* $x_\sigma = \prod$ of her action probabilities along σ.

$$x_\emptyset = 1, \qquad \sum_{a \in A(I)} x_{\sigma_I a} = x_{\sigma_I} \ \ \forall I, \qquad x \geq 0 \qquad \iff \qquad Ex = e,\ x \geq 0 \quad \text{(a "treeplex")}$$

Payoff is *bilinear* in realisation plans: $u(x, y) = x^\top A y$,
$A_{\sigma\tau} = \sum_{\text{leaves } z \text{ reached by } (\sigma,\tau)} \text{chance}(z)\,\text{payoff}(z)$.

Koller–Megiddo–von Stengel (1994/96), von Stengel (1996)

$$\max_{x,\,q} \ f^\top q \quad \text{s.t.} \quad A^\top x - F^\top q \geq 0, \quad Ex = e, \quad x \geq 0$$

— an LP whose size is *linear* in the game tree (the normal form is exponential).

- Dual variables q_J = values of the opponent's information sets J (= counterfactual best-response values).
- So: why not just call an LP solver?

POKER Where the LP breaks: the river spot

Ranges (illustrative single-raised pot):

Big blind: 473 hand combinations; Button: 328.

Bets: 33%, 75%, all-in; raises: all-in.

Sequence form (we computed):

- 7,569 + 5,249 sequences, 4,808 information-set rows;
- A has 2.9 million nonzeros: every leaf contributes a *dense* hand×hand block.

We computed (one core)

HiGHS LP: not finished after 25 min

DCFR, 5,000 iterations: exploitability 1.3×10^{-4} chips in 16 s

The payoff blocks are what LP and sparse direct methods choke on — and what CFR handles with a few matrix–vector products.

MATH Regret matching (Hart & Mas-Colell 2000)

Repeated decision with actions A, payoffs $u_t(a)$ revealed after each round.

$$R_T(a) = \sum_{t=1}^{T} \big(u_t(a) - u_t(\sigma_t)\big), \qquad \sigma_{T+1}(a) = \frac{[R_T(a)]^+}{\sum_b [R_T(b)]^+} \quad \text{(uniform if all } \le 0\text{)}.$$

Blackwell approachability ⇒ no regret

$$\max_a R_T(a) \;\le\; \Delta\sqrt{|A|\,T} \qquad (\Delta = \text{payoff range}).$$

Self-play ⇒ equilibrium ("folk theorem")

If both players have average regret $\le \varepsilon$, the *average* strategies $(\bar{x}_T, \bar{y}_T)$ are a 2ε-equilibrium.

- No step size, no projection, no tuning — just "play actions in proportion to how much you regret not playing them".
- Rate $O(T^{-1/2})$ in theory; as a numerical method: a nonsmooth, parameter-free first-order iteration.

MATH Counterfactual regret minimisation (Zinkevich et al. 2007)

Run regret matching *locally at every information set*, on **counterfactual values**

$$v_i(\sigma, I, a) = \sum_{h \in I} \pi^{\sigma}_{-i}(h) \sum_{z \sqsupset ha} \pi^{\sigma}(ha \to z)\, u_i(z)$$

(opponent and chance reach probabilities only — "as if I had tried to reach I").

Regret decomposition

$$R_T^{\text{full}} \;\le\; \sum_I \left[R_T^{\text{imm}}(I)\right]^+ \quad\Longrightarrow\quad \text{expl}(\bar{x}_T, \bar{y}_T) \;\le\; \frac{\Delta\, |\mathcal{I}| \sqrt{|A|}}{\sqrt{T}}.$$

- One iteration = one pass over the tree: compute reaches top-down, values bottom-up, update regrets.
- Memory: two numbers per (information set, action). Matrix-free: A is never formed.

Vectorised CFR: one pass for all hands

At a public node, store vectors over private hands: reach $\pi_1 \in \mathbb{R}^{n_1}$, $\pi_2 \in \mathbb{R}^{n_2}$. At a leaf:

$$v_1 = \text{amount} \cdot B\,\pi_2, \qquad v_2 = -\text{amount} \cdot B^\top \pi_1, \qquad B_{ij} = \underbrace{C_{ij}}_{\text{no shared card}} \cdot \begin{cases} 1 & \text{fold} \\ \operatorname{sign}(s_i - s_j) & \text{showdown} \end{cases}$$

Structure (s = hand strength, sorted):

- $C = \mathbf{1}\mathbf{1}^\top - K$, K sparse (hands sharing a card: $\approx 8\%$ of pairs);
- $\operatorname{sign}(s_i - s_j)$ = lower-ones minus upper-ones: *semiseparable* (rank-1 off-diagonal blocks).

Consequence

$B\pi$ in $O(n \log n)$ by sorting + prefix sums + a per-card correction, instead of $O(n^2)$ (Johanson et al. 2011).

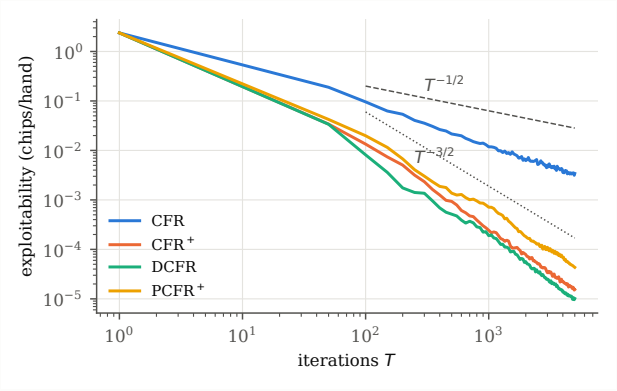
Full river: $n = 1081$ combinations per player $\Rightarrow$ the dense blocks behind the 2.9 M nonzeros.

MATH Modern CFR = a bag of numerical tricks

method	update	numerical reading
CFR$^+$ Tammelin 2014	$R \leftarrow [R + r]^+$; average weighted by t	projection onto the orthant; Cesàro $\rightarrow$ linear averaging
DCFR Brown–Sandholm 2019	positive regrets $\times \frac{t^{1.5}}{t^{1.5}+1}$, negative $\times \frac{1}{2}$; average $\times (\frac{t}{t+1})^2$	forgetting factors: down-weight the transient
PCFR$^+$ Farina et al. 2021	$\sigma_t \propto [R_{t-1} + m_t]^+$, prediction $m_t = r_{t-1}$	optimism / extragradient
alternating	update player 2 using player 1's *new* strategy	Gauss–Seidel vs Jacobi

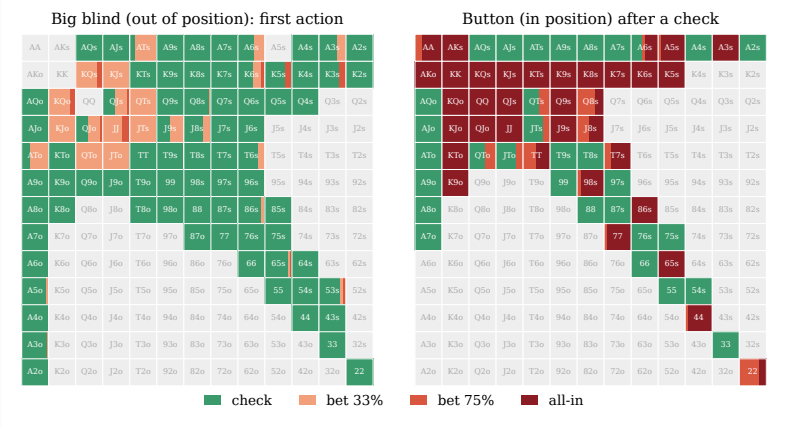
- All are matrix-free and parameter-light; worst-case theory is still $O(T^{-1/2})$.
- In practice: $\approx T^{-1}$ to $T^{-3/2}$ for averages; the *last iterate* oscillates.

MATH We computed: the baselines on Leduc hold'em



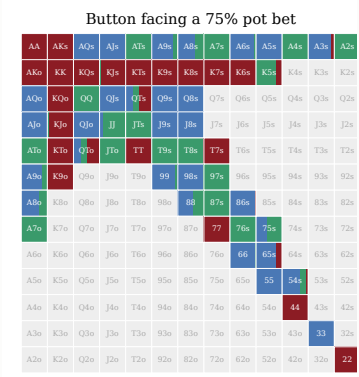
- Our range-vectorised implementation (alternating updates), exploitability of the *average* strategy.
- 5,000 iterations, 14 s: DCFR reaches 1.0×10^{-5}.
- 50,000 iterations, 130 s: CFR^{+} 9.2×10^{-7}, DCFR 1.6×10^{-6}.
- Each extra digit costs $\approx 10\times$ more iterations.

POKER What a solution looks like: first actions



We computed (DCFR, 5,000 iterations, exploitability 1.3×10^{-4} chips). Upper triangle: suited; lower: offsuit; grey: not in range. Each cell shows the action mix, weighted over its combinations.

POKER Facing a bet: the solver rediscovers pot odds



defender	line	s	defends	MDF
Button	BB checks, BTN bets 75, BB jams	0.10	64.7%	90.9%
Button	BB bets 33	0.33	74.3%	75.2%
Button	BB checks, BTN bets 33, BB jams	0.40	57.8%	71.2%
Button	BB bets 75	0.75	55.8%	57.1%
Button	BB bets 100	1.00	49.1%	50.0%
Big blind	BB bets 75, BTN jams	0.10	63.4%	90.9%
Big blind	BB checks, BTN bets 33	0.33	52.2%	75.2%
Big blind	BB bets 33, BTN jams	0.40	56.3%	71.2%
Big blind	BB checks, BTN bets 75	0.75	37.5%	57.1%
Big blind	BB checks, BTN bets 100	1.00	32.6%	50.0%

Button: MDF to within 1.5 points. Big blind *under*-defends: the Button's marginal bluffs have showdown value, so they are indifferent at EV>0, not at 0.

MATH Why push precision further?

Practical solvers stop at ≈ 0.1–0.5% of the pot. Reasons to want more:

- **Subgame re-solving** (DeepStack, Libratus) stitches many solves together; errors compound across streets.
- **Exact mixing frequencies and supports**: which hands *exactly* mix? First-order methods only hint.
- **Certificates**: a primal–dual pair with residual 10^{-13} is a proof of (near-)optimality.
- **Research**: benchmarking algorithms needs ground truth.

The numerical analyst's view

CFR is a first-order method on a nonsmooth, degenerate saddle-point problem: robust, cheap per step, *sublinear* tail. The classical cure for a slow tail is a *local, higher-order* method started from a good point.

NEW Attempt 1 (didn't work): Anderson acceleration of CFR$^+$

- One CFR$^+$ sweep is a map G on the regret vector: $R \mapsto [R + r(\sigma(R))]^+$.
- Fixed points of $G \Leftrightarrow \sigma(R)$ is an equilibrium. Natural idea: accelerate the fixed-point iteration.
- Type-II Anderson acceleration (Anderson 1965; Walker–Ni 2011), memory 10, with safeguards (clip to ≥ 0, restart on residual growth).

We computed

Kuhn: exact equilibrium. Leduc: 2.3×10^{-4} after 2,000 map evaluations — DCFR has 1.9×10^{-4} after 1,000. No gain.

Why: G is piecewise linear and nonsmooth; its fixed points are non-isolated (whole rays, since $\sigma(cR) = \sigma(R)$); AA theory (Toth–Kelley 2015) needs a smooth contraction. We also tried vector extrapolation (RRE) of the averages, and Levenberg–Marquardt on the KKT system: same story.

NEW Reformulate: a monotone variational inequality

Unknowns $z = (x, y, p, q)$: both realisation plans and both dual (value) vectors. Equilibrium $\iff$ LP optimality for both players:

$$0 \le x \;\perp\; E^\top p - Ay \;\ge 0, \qquad Ex = e,$$
$$0 \le y \;\perp\; A^\top x - F^\top q \;\ge 0, \qquad Fy = f.$$

Equivalently, with $w(z) = Mz + b$:

$$M = \begin{pmatrix} 0 & -A & E^\top & 0 \\ A^\top & 0 & 0 & -F^\top \\ -E & 0 & 0 & 0 \\ 0 & F & 0 & 0 \end{pmatrix}, \qquad M + M^\top = 0.$$

Key facts

- M is **skew**: the VI is monotone, the solution set is a polyhedron (non-isolated, cf. Kuhn).
- p, q = information-set values; CFR's best-response values give them for free: $p_I \approx r(h)\,\mathrm{BR}(I, h)/Z$.

NEW Outer loop: proximal point (Rockafellar 1976)

$$z^{k+1} \;=\; \text{solution of the VI with } \; w_\rho(z) = (M + \rho I)\, z + b - \rho\, z^k.$$

- $M + \rho I$ is positive definite ($d^\top (M + \rho I) d = \rho \|d\|^2$) $\Rightarrow$ **unique** solution of each subproblem.
- Converges to a solution of the original VI for any $\rho > 0$.
- Polyhedral problem $\Rightarrow$ **linear** rate, factor $\to 0$ as $\rho \to 0$ (Luque 1984).
- Small ρ: fast outer loop, harder subproblems.

We computed (Leduc, $\rho = 10^{-5}$)

VI residual after each prox step:

$3.4 \times 10^{-2} \to 4.8 \times 10^{-6} \to 3.3 \times 10^{-9}$
$\to 6.9 \times 10^{-12} \to 2.0 \times 10^{-14}$

$\approx$ three digits per step.

NEW Inner loop: semismooth Newton on Fischer–Burmeister

Replace complementarity $0 \le a \perp b \ge 0$ by $\varphi(a,b) = a + b - \sqrt{a^2 + b^2} = 0$ (Fischer 1992):

$$\Phi(z) = \big(\varphi(x_i, w_i)\big)_{\text{bounded}}, \quad \big(w_j\big)_{\text{free}} = 0, \qquad V = D_a + D_b\,(M + \rho I) \;\in\; \partial\Phi(z),$$

$D_a, D_b \ge 0$ diagonal with $(D_a)_{ii} + (D_b)_{ii} \ge 2 - \sqrt{2}$.

Lemma. Every such V is nonsingular.

If $Vd = 0$, put $u = (M + \rho I)d$. Row i: $(D_a)_{ii} d_i + (D_b)_{ii} u_i = 0$ with nonnegative, not both zero coefficients $\Rightarrow d_i u_i \le 0$. Hence $\rho\|d\|^2 = d^\top u = \sum_i d_i u_i \le 0$, so $d = 0$. □

- $\theta = \frac{1}{2}\|\Phi\|^2$ is C^1 $\Rightarrow$ Armijo line search; for strongly monotone problems every stationary point of θ is the solution (De Luca–Facchinei–Kanzow 1996).
- Globally convergent, locally **quadratic** (strong semismoothness; Qi–Sun 1993).

POKER NEW Poker makes the Newton step cheaper

- Rows with $(D_b)_{ii} \approx 0$ ($x_i = 0$ clearly optimal: action dropped) are identity rows: **eliminate** them.
- What remains: current support + information sets.
- Equilibria in poker are *mostly pure*: most hands take one action.

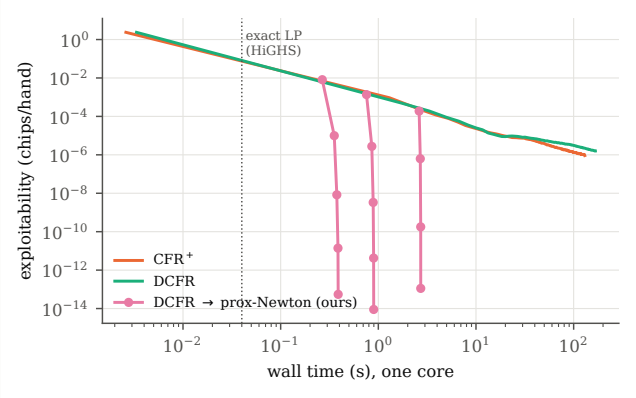
We computed (exact equilibria)

	Leduc	small river
pure (infoset, hand)	49%	71%
sequences in support	59%	52%
Newton system size	964	2,864

The hybrid solver

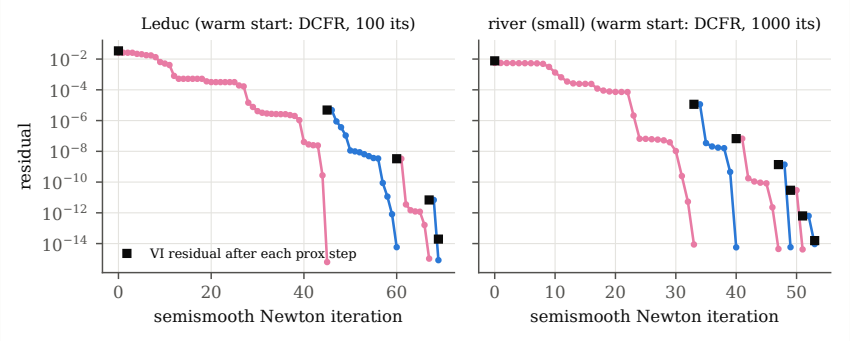
1. DCFR for k iterations (cheap, global) $\rightarrow$ average strategy $(\bar{x}, \bar{y})$;
2. one best-response pass $\rightarrow$ duals (p, q);
3. proximal semismooth Newton ($\rho = 10^{-5}$) until the VI residual is $\leq 10^{-13}$.

NEW We computed: Leduc hold'em



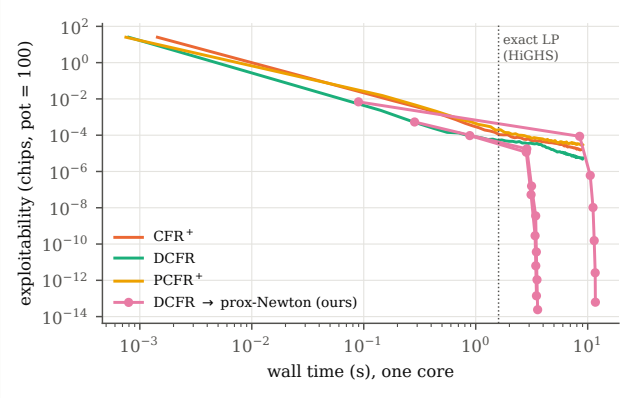
- DCFR 100 iterations + 4 prox steps (70 Newton steps): exploitability 5.5×10^{-14} in 0.39 s.
- CFR$^+$ after 130 s: 9.2×10^{-7}.
- Exact LP (HiGHS): 0.04 s. At this size, the LP wins — the point is the *shape* of the curves.

NEW Inside the finish: identify, then terminate



- Each colour = one proximal subproblem. Long plateau: Newton is *identifying the active set* (damped steps).
- Then the residual collapses in 1–3 steps: on a piecewise-linear problem, Newton with the right pieces is exact.
- Later subproblems start with the right active set: 2–7 Newton steps each.

NEW We computed: a real river spot (small ranges)



Board K♠7♢2♣T♡4♠, 147 vs 139 combos, bet 75% / all-in.

- Hybrid: 2.4×10^{-14} chips in 3.6 s (54 Newton steps).
- DCFR: 5.4×10^{-6} after 30,000 iterations (9 s).
- HiGHS LP: 1.6 s.

NEW Where it does not (yet) win: full-size ranges

Medium spot (473 vs 328 combos, 3 bet sizes):

- DCFR iteration: 3 ms.
- One Newton step = one sparse LU of a 17,626 × 17,626 matrix with 5.9 M nonzeros: 25 s, 55 M nonzeros of fill (COLAMD).
- The dense hand×hand payoff blocks fill in.

Verdict

CFR for the first digits, at scale. Newton for the last digits, while the (support-reduced) Newton system can be factored.

How to scale it (open):

- Newton–Krylov: products with A cost one tree pass, $O(n \log n)$ per leaf by the semiseparable+sparse structure.
- Needs a preconditioner for $D_a + D_b(\rho I + \text{skew})$: hierarchical (HSS) factorisation of the payoff blocks?
- Better active-set prediction from the CFR iterate ⇒ smaller systems.

NEW What is new here, and what is not

- **Not new**: sequence-form LP (von Stengel 1996); proximal point (Rockafellar 1976); FB semismooth Newton (Fischer 1992; De Luca–Facchinei–Kanzow 1996); CFR family.
- **Closest work**: Yang, Gao, Lin & Kroer (arXiv 2512.12910, 2025): semismooth Newton for *matrix* games via Douglas–Rachford splitting, warm-started by PRM$^+$; experiments on normal-form games (Kuhn as a 27×64 matrix).
- **Here**:
  - Newton directly on the *sequence-form* complementarity system of extensive-form poker;
  - skew-monotone VI + proximal regularisation $\Rightarrow$ uniformly nonsingular Newton matrices (one-line proof);
  - CFR iterate + best-response values as a primal–dual warm start;
  - tested on Leduc and no-limit river subgames with card removal.
- To our knowledge no poker solver uses a second-order finish.

Takeaways

1. A poker equilibrium is the solution of a structured bilinear saddle-point problem; the sequence form makes it an LP of size linear in the tree.
2. CFR = regret matching at every information set: a matrix-free, nonsmooth first-order method with a provable $O(T^{-1/2})$ rate and a much better practical one — plus numerical tricks (projection, discounting, optimism, Gauss–Seidel).
3. The solver's output is readable poker: polarised betting, block bets, pot-odds defence (to within 1.5%).
4. Read the equilibrium conditions as a **skew-monotone VI**: proximal point + Fischer–Burmeister semismooth Newton turns CFR's slow tail into a few linear solves and reaches 10^{-13}.
5. Scaling the finish is a structured numerical linear algebra problem (semiseparable + sparse blocks).

Code: `github.com/dwangnumbers/Poker-Solver` — solver, benchmarks, figures, these slides.

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Backup: implementation summary

`games.py`	Kuhn, Leduc, no-limit river builder (7-card evaluator, card removal, range notation)
`cfr.py`	range-vectorised public-tree CFR / CFR$^+$ / DCFR / PCFR$^+$, alternating updates, best response, exploitability
`seqform.py`	sequence form (E, F, A), dual estimates from best responses, prox semismooth Newton (FB, reduced)
`lp.py`	reference LP via HiGHS
`aa.py`, `extrap.py`	the negative results (Anderson, RRE)
`bench_*.py`, `fig_*.py`	every number and figure in these slides

- Exploitability: $\frac{1}{2}(\mathrm{BR}_1 + \mathrm{BR}_2)$ by exact best-response passes, normalised by the deal weight Z.
- Sequence form checked against the LP: Leduc value -0.0856064240780 (both).
- Dual warm start: $p_I = r_1(h)\,\mathrm{BR}_1(I,h)/Z$, $q_J = -r_2(h)\,\mathrm{BR}_2(J,h)/Z$.

Backup: effect of the proximal parameter

We computed (Leduc, warm start DCFR 200 iterations, final exploitability $\leq 10^{-13}$ in all cases):

ρ	10^{-3}	10^{-4}	10^{-5}	10^{-6}
prox steps	19	6	4	3

Continuation ($\rho_k \downarrow$) did not reduce the total number of Newton steps on the small river spot (54 fixed vs 63–96).